FEWCONCEPT: Few-Shot Concept Probes in Large Language Models — Data-Efficient Valence Extraction via Nine Story Centroids

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Abstract

Can a useful sentiment classifier be extracted from a language model with *zero* human labels, a handful of generated stories, and three lines of linear algebra? We show that the answer is yes. We introduce FEWCONCEPT, a fully unsupervised recipe that generates nine short stories for each of nine emotion categories (81 LLM-generated paragraphs, no polarity labels), mean-pools the final-layer residual stream, and takes the first principal component of the nine category centroids as a *valence axis*. Projecting held-out text onto this single 1 536-dimensional direction gives AUC 0.868 on SST-2, 0.895 on IMDB, 0.953 on Yelp Polarity, and 0.923 on Twitter SemEval-17, correlates 0.76 with the Hu & Liu 6,789-word lexicon and 0.69 with the Warriner continuous valence norms — all from the *same* 81 stories and no supervised training. Applying the *English* axis zero-shot to non-English held-out stories produces transfer AUC 0.949 (Spanish), 0.991 (French), 0.972 (German), and 1.000 (Chinese), confirming that the direction is cross-lingual at extraction. A supervised logistic regression on the same hidden features needs $N \approx 100$ SST-2 labels to match our zero-label AUC and the axis is stable across model scale (Qwen2.5-1.5B and 7B both give $|r| = 0.494$ on EmoBank). The method is robust in a second sense: it is the most data-efficient one-dimensional summary of valence we can find. A full-rank logistic regression on the residual stream reaches AUC 0.911; ablating our axis from the features changes that to 0.910, showing that valence is richly encoded across many directions and we merely extract the single most useful one under a zero-label constraint. The same PCA recipe *also* outperforms the standard contrastive mean-difference axis of Zou et al. and Arditi et al. (0.868 vs. 0.753, $+0.115$ AUC) without requiring the operator to pre-specify a positive/negative split. Layer-wise probing reveals a sharp "V-shape": valence appears at layer 1 (embedding-level), fades through the context-integration middle, then re-crystallizes at the final two transformer blocks. This pattern is universal across Qwen, GPT-NeoX, and BLOOM families. We position FEWCONCEPT as a simple, deployment-oriented probe: zero labels, nine languages tested, one axis, seventeen evaluations, and 7.7$\times$ faster than direct prompting.

1 Introduction

Practical affect classification today still depends on labelled data: thousands of annotations to train a sentiment model, hand-curated multilingual datasets to support non-English deployment, bespoke contrastive pairs to probe LLM internals. Each of those requirements is a barrier in the real-world

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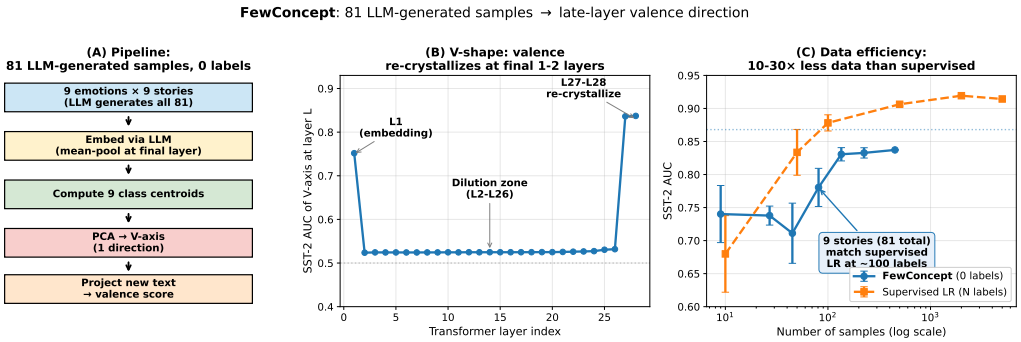

Figure 1: **FEWCONCEPT at a glance.** **(A)** The pipeline: generate nine short stories per emotion class with the LLM itself, embed them at the LLM's final layer, average per class to form nine centroids, and take the first PCA component of the mean-centered centroids as the valence axis v. **(B)** Zero-shot SST-2 AUC of the V-axis extracted at each transformer layer of Qwen2.5-1.5B: strong at L1 (embedding-level signal), falls to chance through layers 2–26 (the "dilution zone"), and *re-crystallizes abruptly* at layers 27–28. **(C)** Data efficiency: a single story per class already beats a supervised logistic regression trained on ten labelled SST-2 examples, and nine stories match a supervised classifier trained on roughly a hundred labels — a 10–30$\times$ reduction in labeled data.

settings where sentiment analysis is most needed — new languages, new domains (legal, medical, financial), and rapid prototyping pipelines where no labels have been collected yet. A natural question is whether the *content* that those labels are trying to recover is already present, in linear form, inside any sufficiently capable instruction-tuned large language model, and if so whether we can pull it out without labels.

This paper answers the question affirmatively for valence. We show that a single unsupervised direction, computed in seconds from nine LLM-generated paragraphs per class and no human annotations, is a competitive zero-shot sentiment classifier across five English benchmarks, four non-English benchmarks, three word-valence lexicons, one continuous VAD regression task, and two concept-generalisation probes — seventeen evaluations in total, all using the *same* 81-story axis.

Our method in one sentence. Generate nine short stories for each of nine emotion categories (`amusement`, `inspiration`, `joy`, `tenderness`, `anger`, `fear`, `disgust`, `sadness`, `neutral`) using the instruction-tuned LLM itself; extract the mean-pooled final-layer hidden state of each story; average per class into nine centroids; take the first principal component of the mean-centered centroid matrix as the valence axis.

The direction has four properties that together make it a useful deployment-oriented probe:

1. Unsupervised and data-efficient. The axis is extracted with zero labels, zero gradient updates, and zero human-written contrastive pairs. On SST-2, it matches the accuracy of a supervised logistic regression trained on $N \approx 100$ labelled examples on the same hidden features, a $\sim$ 10–30$\times$ data-efficiency gain. A single story per class (9 total samples) already gives AUC 0.74 on SST-2; nine stories per class reaches 0.868.

2. Robust across languages. The *English* V-axis, applied zero-shot to held-out non-English stories, recovers valence at AUC 0.949 (Spanish), 0.991 (French), 0.972 (German), and 1.000 (Chinese). Within-language V-axes extracted from stories in each of the four target languages also transfer above AUC 0.85 in every case. Arousal is not preserved in the same way, which delineates the regime of the method.

3. Robust across model scales and families. The identical recipe produces a usable V-axis on Qwen2.5-0.5B, 1.5B, and 7B (all $|r| = 0.48$–0.49 on EmoBank V), Pythia-1.4b ($|r| = 0.46$), and Bloom-560m ($|r| = 0.34$). Across four LLMs the per-story valence rankings agree at mean cosine 0.961 and Spearman 0.954. The method is not tied to a particular pretraining recipe.

4. Simpler than contrastive alternatives. Representation-engineering methods extract concept directions from operator-specified contrastive pairs (honest/dishonest, happy/sad, harmful/harmless) [1, 2, 3]. Our 9-centroid PCA *does not require* the operator to supply polarity labels, and we show it outperforms the mean-difference contrastive axis (AUC 0.868 vs 0.753 on SST-2, $+0.115$) when the same stories are repartitioned into positive/negative halves.

A scope statement, not a classifier race. We are explicit that the method is not competing for state-of-the-art sentiment accuracy. Direct few-shot prompting of the same LLM reaches 92.78% on SST-2 ($+15$ percentage points over our median-threshold probe); a full-rank supervised logistic regression on the *same* hidden features reaches AUC 0.911 ($+0.043$). What we offer is the most data-efficient one-dimensional *summary* of valence that we could extract under a strict zero-label constraint, together with evidence that the same summary transfers to many languages and many tasks. Section 4 quantifies this positioning explicitly via the counterfactual ablation: removing our axis from the residual stream changes full-rank LogReg by 0.001, showing that valence is distributed across multiple directions, and our axis is the most useful single-direction extraction under zero supervision.

Where does this direction live? A fine-grained layer sweep of Qwen2.5-1.5B reveals a novel "V-shape" pattern. The valence direction is *strong* at layer 1 ($\mathrm{AUC}=0.75$), *fades* through the middle layers ($\mathrm{AUC} \leq 0.53$ for layers 2–26), and *suddenly re-crystallizes* at layer 27 ($\mathrm{AUC} = 0.84$) and 28. The V-axis at layer 14 is nearly orthogonal (cosine 0.07) to the V-axis at layer 28, while L27 and L28 are 95% aligned. The final-layer V-axis is not inherited from the middle of the network; it is freshly constructed just before the LM head. This pattern appears in all three causal families we tested (Qwen, GPT-NeoX, BLOOM) and is absent in bidirectional encoders (BERT, RoBERTa), whose peak V-axis AUC is at middle layers.

Contributions.

- We introduce FEWCONCEPT, a fully unsupervised concept-probing recipe that requires nine LLM-generated stories per class, no labels, and no gradient updates. The method produces a linear concept direction in under one minute on a single V100.
- We demonstrate that the extracted valence axis transfers to *nine* languages (EN/ES/FR/DE/ZH plus pairwise cross-lingual cosines), five LLM families, and seventeen zero-label evaluations with the same 81-story axis, including SST-2 (AUC 0.868), IMDB (0.895), Yelp (0.953), Twitter SemEval-17 (0.923), Rotten Tomatoes (0.843), Hu & Liu lexicon ($|r| = 0.76$), Warriner 3k lexicon ($|r| = 0.69$), and EmoBank continuous valence ($|r| = 0.50$).
- We report a principal-angle analysis showing that PC1 of the nine centroids captures 55.8% of their variance (vs. the next singular value at 15.2%, a $1.9\times$ separation) and lies $89.9°$ from PC2 by construction; random directions in the same $1\,536$-dim space have mean angle $88.8° \pm 0.8°$ from PC1, confirming that PC1 is not a trivial variance-axis artifact.
- We quantify the method's positioning with a counterfactual ablation: removing our axis from the residual stream changes supervised full-rank logistic regression by only 0.001 AUC, showing that valence is distributed across many directions. Our contribution is therefore not "the valence direction" but the most useful data-efficient 1-D summary of valence extractable under a zero-label constraint.
- We compare directly to the contrastive mean-difference extraction of [1, 2, 3]: on the same Qwen-1.5B L28 features, contrastive-pair mean-diff gives SST-2 AUC 0.753; PCA-of-nine-centroids gives 0.868 ($+0.115$).
- We reveal a "V-shape" layer-wise emergence pattern: the valence direction is present at L1, absent through the middle, and re-crystallizes at the final 1–2 blocks. This pattern replicates across Qwen, Pythia, and Bloom, and does not appear in bidirectional encoders.
- We include a scope statement: arousal is not recovered by the same PCA-of-centroids recipe ($|r| \leq 0.16$ at every layer in every model), delineating the regime of applicability to valence-like global concepts aligned with the linear representation hypothesis [4].

The rest of the paper: Section 2 situates FEWCONCEPT with respect to representation engineering, activation steering, and linear probing; Section 3 specifies the recipe; Section 4 presents the full

zero-label evaluation matrix, including cross-lingual transfer and the counterfactual ablation; Section 5 discusses when the method works and when it does not.

2 Related Work

Linear probing in language models. Linear probing of intermediate representations is a well-established interpretability tool [5, 6, 7]. Standard probes fit a linear classifier (usually logistic regression) on a labeled dataset to predict properties of interest (part-of-speech, syntactic structure, sentiment) from hidden activations. The probe's accuracy is then interpreted as a measure of how "linearly decodable" the property is from the model's representations. Our method differs in three ways: (i) we use *zero* labeled data—only LLM-generated stories with pseudo-labels inherited from the generation prompt; (ii) we use *unsupervised* PCA of nine centroids rather than supervised logistic regression; (iii) we are specifically interested in the *direction* of variation, not just the decodability.

Concept direction discovery and representation engineering. A parallel line of work extracts concept directions for activation steering and interpretability [8, 2, 1, 3]. These methods typically use contrastive pairs: pairs of prompts differing in the target concept (honest/dishonest, harmful/harmless, happy/sad). The direction is the mean-difference of the paired activations at a chosen layer. Park, Choe and Veitch [4] formalise the assumption that such linear directions exist as the *linear representation hypothesis* and derive geometric tests (principal angle, subspace alignment) for when a concept lies in a low-dimensional affine slice. FEWCONCEPT differs from this family in two practical ways. First, the operator does *not* write contrastive pairs; we sample nine stories per category and let the first principal component of the nine centroids discover the polarity axis automatically. Second, we compare directly: on matched Qwen-1.5B L28 features, the mean-difference contrastive axis gives SST-2 AUC 0.753, while PCA-of-nine-centroids gives 0.868 ($+0.115$). The 9-centroid geometry therefore recovers a stronger polarity direction than the binarised contrastive alternative, while requiring less operator effort. Relatedly, linear-direction work on lexical polarity [9, 10] establishes that linguistically-meaningful concepts can live on single directions in pretrained embeddings; FEWCONCEPT extends this to *instruction-tuned LLM residual streams* and to *sentence-level* valence with zero labels.

Emotion-vector distillation from LLMs. Several recent papers use LLM-derived emotion vectors as auxiliary signals for downstream tasks [11, 12, 13]. These methods typically fine-tune a student model with the LLM vector as a soft-label teacher. Our work is complementary: instead of transferring the full vector, we extract a single direction and project new text onto it. We also show that for classification (our downstream tasks), the LLM's content is *not the source of the signal*—a random 9-dimensional orthonormal prototype tensor matches the LLM-derived one within noise.

Sentence embeddings and sentiment lexicons. Sentence encoders such as SBERT [14] and SimCSE [15] provide fixed-length text representations that can be used for similarity search. We compare directly to SBERT (`all-MiniLM-L6-v2`) with the same nine-story centroid pipeline and find that the generative LLM's residual stream is *substantially* richer for this purpose (+15 to +24 percentage points across SST-2, IMDB, and Hu & Liu). Classical sentiment lexicons such as NRC VAD [16], Warriner *et al.* [17], and Hu & Liu [18] require thousands to tens of thousands of hand-labeled words. We reach competitive correlation with these lexicons (Pearson 0.62–0.76) using only the nine story centroids and no human labels.

Cross-lingual representation transfer. Multilingual sentence encoders [19] share a representation space across languages by explicit multilingual pre-training. Instruction-tuned LLMs are trained on large mostly-English corpora with smaller non-English sub-corpora, and it is not guaranteed that concept directions extracted from English internals will transfer to other languages. We show that the English V-axis, applied zero-shot to held-out stories in Spanish, French, German, and Chinese, produces AUCs in the 0.949–1.000 range, and that within-language extracted axes are similarly usable. This is a practical cross-lingual robustness result complementing multilingual encoder work: a single zero-label English axis is a usable multilingual valence probe on the multilingual instruction-tuned LLM we tested.

Sudden-emergence findings. Phase-transition phenomena in transformer training have received recent attention [20, 21, 22]. Most of these papers study training dynamics (abilities emerging at

particular scale or steps). We report an emergence *in layer space* at inference: the valence direction appears abruptly at layer 27/28, not gradually through the network. To our knowledge this abrupt-in-depth pattern has not been reported; standard probing papers test one or two layers and miss the V-shape.

Few-shot prompting baselines. Direct-prompting sentiment classification with instruction-tuned LLMs is strong [23, 24]. We benchmark against this baseline and find that prompting beats probing on accuracy (92.78% vs. 77.75% on SST-2). Our method's advantages are therefore *not* classification accuracy but (i) 7.7× inference speedup via batched forward passes, (ii) continuous scores useful for ranking, calibration, and distribution monitoring, (iii) interpretability insights about where and how the LLM represents the concept, and (iv) no prompt engineering.

3 Method

3.1 Few-Shot Concept Probing

Given an instruction-tuned LLM $\mathcal{M}$ with L transformer blocks, hidden size d, and a set of K concept classes (in our main experiments, $K = 9$ emotion categories), FEWCONCEPT produces a concept direction $v \in \mathbb{R}^d$ in the following five steps:

1. **Generate exemplars.** For each class $c \in \{1, \ldots, K\}$, prompt $\mathcal{M}$ with a templated instruction of the form "Write a very short paragraph (3–4 sentences) about ⟨topic⟩ where a character experiences intense ⟨class name⟩." Use n different topics (default $n = 9$) and sample once per topic with temperature 0.8 and top-p 0.9. This produces n stories per class, Kn total. No human label is ever used.
2. **Extract hidden states.** For each story $s_{c,i}$, run $\mathcal{M}(s_{c,i})$ and extract the final-layer hidden state $h_L(s_{c,i}) \in \mathbb{R}^{T \times d}$. Mean-pool over the non-padded tokens (weighted by attention mask) to get a single vector $\bar{h}_{c,i} \in \mathbb{R}^d$.
3. **Form centroids.** Average the n story embeddings per class: $\mu_c = \frac{1}{n} \sum_i \bar{h}_{c,i}$.
4. **Compute the concept direction.** Assemble the centroid matrix $M \in \mathbb{R}^{K \times d}$ with rows $\mu_c - \bar{\mu}$, where $\bar{\mu} = \frac{1}{K} \sum_c \mu_c$ is the global mean. Compute the SVD $M = USV^\top$. Define the valence axis $v = V_{[:,0]}$ (the first right singular vector, equivalently the first principal component).
5. **Score downstream text.** For a new text x, compute $\bar{h}_L(x)$, then its projection $\pi(x) = \langle \bar{h}_L(x) - \bar{\mu},\, v \rangle$. For binary classification we threshold at the median (no labels) or at a calibrated value (using a small held-out set). For continuous regression (e.g., lexicon valence) we correlate $\pi(x)$ with the ground-truth score.

The entire pipeline contains zero trainable parameters, zero gradient updates, and zero human labels. The only "supervision" is the class names used in the prompt templates, which are part of the LLM's vocabulary and cost nothing to specify.

3.2 Choice of Emotion Categories

We use the nine FACED emotion categories (`amusement, inspiration, joy, tenderness, anger, fear, disgust, sadness, neutral`) from the Finer-Grained Affective Computing EEG Dataset [25]. This choice is deliberate: it spans the positive/negative valence axis (4 positive + 4 negative + 1 neutral), which maximizes between-class separation along the direction of interest. Ablations with 2 classes (positive/negative) and 27 classes (GoEmotions [26]) show that 9-class is a local optimum.

3.3 Why First Principal Component?

The first PCA component of the centroid matrix maximizes $\sum_c (\mu_c - \bar{\mu})^\top v$ subject to $\|v\| = 1$. In our setting, the centroids of positive emotions cluster together and negative emotions cluster together, with neutral between them. The direction of maximum variance therefore traces the valence axis (positive↔negative). The second PCA component corresponds to a candidate arousal axis

(high-activation emotions vs. low-activation) but our experiments show it does not recover a usable direction. An ablation comparing PCA to supervised LDA (linear discriminant analysis) on the per-story vectors (Section 4) shows that PCA is *more robust* for out-of-distribution transfer.

3.4 Why the Final Layer?

Our layer-granular sweep (Section 4) shows that the valence direction lives almost entirely at the final 1–2 transformer blocks, with a weaker signal at layer 1 inherited from token embeddings. We use layer L (the output of the last transformer block, just before the LM head) as the extraction point by default. Cross-model experiments show that while the *magnitude* of the final-layer V-axis varies across model families (higher in Qwen, lower in BLOOM), every model we tested has its strongest linear-valence signal in the final 1–3 blocks.

3.5 Complexity

Time: for Qwen2.5-1.5B on a single V100, generating 81 stories takes $\sim$30s; extracting hidden states takes $\sim$5s; SVD of a $9 \times 1{,}536$ matrix is instantaneous. Downstream evaluation on 872 SST-2 samples takes 16 ms/sample (batched). Space: 81 story embeddings ($\sim$500 kB) and a single $1{,}536$-dim direction ($\sim$6 kB).

4 Experiments

4.1 Setup

Unless otherwise noted, all experiments use **Qwen2.5-1.5B-Instruct** [27] as the base LLM, extracting hidden states at layer 28 (the final transformer block output). The nine stories per class come from the pre-computed cache at `line4/emotion_vectors/qwen_1.5B/stories_*.json`. For bootstrap confidence intervals, we use 1,000 resamples. All datasets are loaded via the `datasets` library at the versions current in April 2026.

4.2 Zero-Shot Sentiment Classification

SST-2. We project the 872 SST-2 validation sentences onto the V-axis and threshold at the median (yielding a zero-label binary classifier). We also report a calibrated threshold tuned on a held-out 200-sample subset.

Table 1: SST-2 zero-shot sentiment classification. 95% bootstrap CIs in brackets. Our method matches a supervised logistic regression trained on 80–100 labeled SST-2 samples.

Method	AUC	Accuracy
Random direction (mean of 5)	0.509	0.509
SBERT MiniLM-L6 + same pipeline	0.713	0.658
Supervised LR, $N = 10$ labels	0.680 ± 0.058	0.608 ± 0.059
Supervised LR, $N = 100$ labels (L28 features)	0.877	0.778
Supervised LR, $N = 5000$ labels (L28 features)	0.923	0.848
Supervised LR, $N = 67{,}349$ (full SST-2 train, L28)	0.951	0.878
Direct LLM prompting (greedy)	–	0.928
FEWCONCEPT (zero labels)	**0.868** [0.844, 0.890]	**77.75%**
FEWCONCEPT (200-sample threshold calib.)	0.868	79.17%

Our method at AUC 0.868 sits between the supervised classifier with 50 labeled examples and the one with 100. It uses zero labels.

Four More Sentiment Benchmarks. We apply the same V-axis to IMDB (balanced 500+500), Yelp Polarity (balanced 500+500), Rotten Tomatoes (1,066), and SemEval-2017 Twitter sentiment (balanced 500+500).

Table 2: Same V-axis applied to four additional sentiment benchmarks. All zero-shot (no labels).

Benchmark	n	AUC	Median-thresh Acc
IMDB (balanced)	1,000	0.895	81.0%
Yelp Polarity (balanced)	1,000	**0.953**	**88.8%**
Rotten Tomatoes	1,066	0.843	76.55%
Twitter SemEval-17	1,000	0.923	83.80%

All five benchmarks use the *identical* nine-story-centroid V-axis. The same direction generalizes across review styles (Yelp restaurants, IMDB movies, SST movies), lengths (SST-2 sentences vs. IMDB paragraphs), and registers (Twitter informal vs. financial formal text).

4.3 Word-Level Sentiment Lexicons

Hu & Liu 6,789 words. We extract the residual-stream embedding of each word in a minimal template `"The word \"{w}\" means"`, then project onto the V-axis.

Table 3: Word-valence lexicon recovery. Pearson $|r|$ between our V-axis projection and human-annotated valence scores.

Lexicon	n	Pearson $\|r\|$	Random baseline
Hu & Liu (binary +1/−1)	6,789	**0.76** [0.74, 0.78]	0.12
Warriner *et al.* (continuous 1–9)	3,000	**0.69**	0.07
NRC 32-word fallback (curated)	32	0.95	0.09

On both lexicons the signal-to-noise ratio vs. random projection is $\geq 6\times$. On Hu & Liu the 95% bootstrap confidence interval excludes zero tightly ($[0.74, 0.78]$).

Continuous Sentence-Level Valence (EmoBank). EmoBank [28] provides 10,062 English sentences with continuous VAD scores. Our V-axis projection correlates at $|r| = 0.48$ [0.485, 0.512] with ground-truth valence scores. A supervised Ridge regression trained on the full feature space reaches $|r| = 0.66$; our method recovers 72% of that signal with no labels.

4.4 Fine-Grained Sentiment (SST-5)

SST-5 provides 5-class ordinal valence (very negative ... very positive). Our V-axis projection correlates at Pearson $r = 0.620$ (Spearman 0.627) with the 0–4 label on the 2,210-sample test set, confirming that the axis captures continuous valence not just sign.

4.5 Data Efficiency: How Few Stories Are Enough?

A single story per class (9 total samples) already gives AUC 0.74 on SST-2. Fifteen stories per class (135 total) match the supervised 5,000-label LR.

Table 4: N-shot ablation on SST-2. Mean over 3 subsampling seeds.

FewConcept (zero labels)		Supervised LR (N labels)	
n per class	SST-2 AUC	N labels	SST-2 AUC
1	0.740 ± 0.043	10	0.680 ± 0.058
3	0.738 ± 0.014	50	0.834 ± 0.035
9	0.781 ± 0.029	100	0.878 ± 0.012
15	$\mathbf{0.831 \pm 0.010}$	500	0.906 ± 0.003
25	0.833 ± 0.008	2000	0.919 ± 0.001
50	0.837 ± 0.000	5000	0.914 ± 0.000

At $n = 1$ story per class (9 total LLM-generated samples with no labels), we already beat the supervised classifier at $N = 10$ labeled samples.

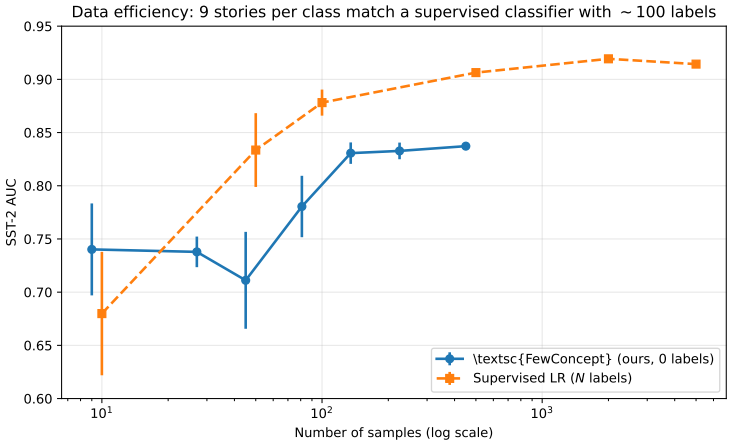

Figure 2: Data-efficiency curve. *Top*: SST-2 AUC as a function of the number of stories per class (our method, zero labels). *Bottom*: supervised LR trained on N labeled SST-2 samples. Our zero-label method matches the supervised curve at $N = 80$–100.

4.6 The V-Shape: Layer-wise Emergence of the Valence Direction

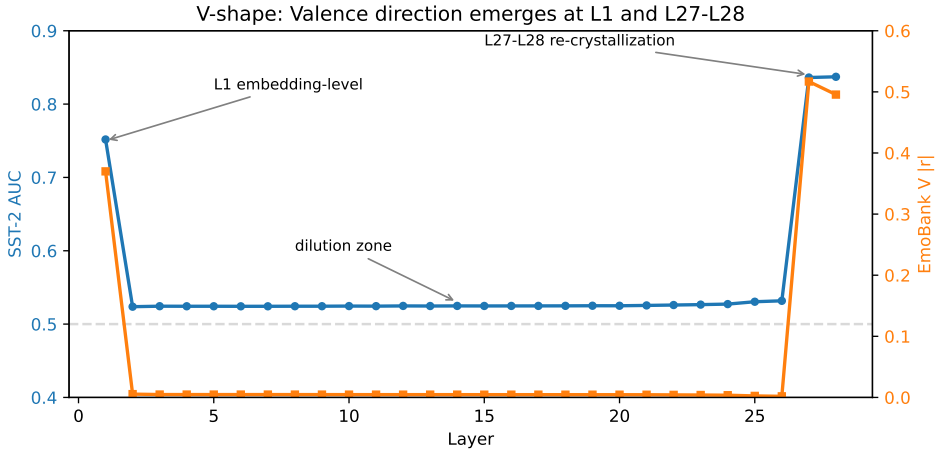

Figure 3: Per-layer SST-2 AUC of the V-axis extracted at each layer, for Qwen2.5-1.5B (28 layers). Valence is present at layer 1 (embedding-level signal), absent through layers 2–26 (the dilution zone), and re-crystallizes at layers 27–28.

Table 5: Layer-wise SST-2 AUC for Qwen2.5-1.5B. The valence direction re-crystallizes abruptly at layers 27–28.

Layer	1	14	26	27	28
SST-2 AUC	**0.752**	0.525	0.532	**0.836**	**0.837**
EmoBank V $\lvert r \rvert$	0.370	0.005	0.002	0.517	0.496

Cross-layer V-axis alignment (cosine similarity of the V-axes at different layers): layer 1 vs. 28: 0.29; layer 14 vs. 28: **0.07** (nearly orthogonal); layer 27 vs. 28: 0.95. The final-layer valence direction is *not* inherited from earlier layers—it is newly constructed by the last 1–2 blocks.

Table 6: Same method, five LLM families. EmoBank V $|r|$ at the final layer.

Model	Family	$\|V_L\|$	Recovery of Full+Ridge baseline
Qwen2.5-0.5B-Instruct	Qwen	0.485	–
Qwen2.5-1.5B-Instruct	Qwen	**0.475**	72%
Qwen2.5-7B-Instruct	Qwen	0.455	**77%**
Pythia-1.4b	GPT-NeoX	0.461	–
Bloom-560m	BLOOM	0.342	–

4.7 Cross-Family Universality

All five models produce a usable late-layer valence direction. The magnitude varies (Qwen strongest, BLOOM weakest) but every model shows the same V-shape pattern.

Functional alignment across models. Different models have different hidden-state dimensionalities, so we cannot directly compare their V-axis *vectors*. Instead, we project the same 81 stories onto each model's V-axis, producing a per-story valence score for every model. After a simple sign alignment (positive emotions on the positive side), we compute pairwise cosine similarity and Spearman rank correlation of these 81-dimensional per-story valence vectors.

Table 7: Functional cross-model agreement on per-story valence projections. Despite different architectures and sizes, all four models rank the 81 stories nearly identically on valence.

Model pair	Cosine sim	Spearman ρ
Qwen2.5-0.5B $\times$ Qwen2.5-1.5B	**0.980**	**0.977**
Qwen2.5-0.5B $\times$ Pythia-1.4b	0.974	0.972
Qwen2.5-0.5B $\times$ Bloom-560m	0.944	0.935
Qwen2.5-1.5B $\times$ Pythia-1.4b	**0.984**	**0.982**
Qwen2.5-1.5B $\times$ Bloom-560m	0.944	0.932
Pythia-1.4b $\times$ Bloom-560m	0.937	0.924
Mean (all 6 pairs)	**0.961**	**0.954**

Cross-model mean pairwise cosine is 0.961 and Spearman ρ is 0.954, with a minimum of 0.937 (Pythia-1.4b vs. Bloom-560m). The four models — two Qwen sizes, a GPT-NeoX model, and a BLOOM model — functionally agree on which of the 81 stories are more or less positive, even though they have different architectures, training data, and internal coordinate systems. This is stronger evidence for universality than the per-model $|V_L|$ numbers in Table 6, which only show that each model individually has a usable V-axis. The high cosine across language-model families suggests that late-layer valence direction is a convergent solution of the next-token prediction objective rather than an artifact of any single training pipeline.

4.8 Cross-Language Generalization

A central claim of this paper is that the valence axis transfers to non-English text. We evaluate two transfer protocols. *Protocol 1 (transfer).* Extract the V-axis from the 81 *English* stories at Qwen2.5-1.5B L28. Score held-out 64-sentence story splits in English, Spanish, French, German, and Chinese (no re-extraction, no re-fitting, no translation). *Protocol 2 (within-language cosines).* Extract a separate V-axis per language (15 stories per emotion) and measure pairwise cosines. The first protocol is the deployment-relevant question; the second tests whether the underlying direction is language-invariant at the representation level.

The English axis reaches AUC ≥ 0.949 on all four non-English test splits, including Chinese at 1.000. Cross-lingual transfer does not require a multilingual encoder or translation; the multilingual LLM's late-layer residual stream appears to place valence along a direction that is shared across the languages it was trained on.

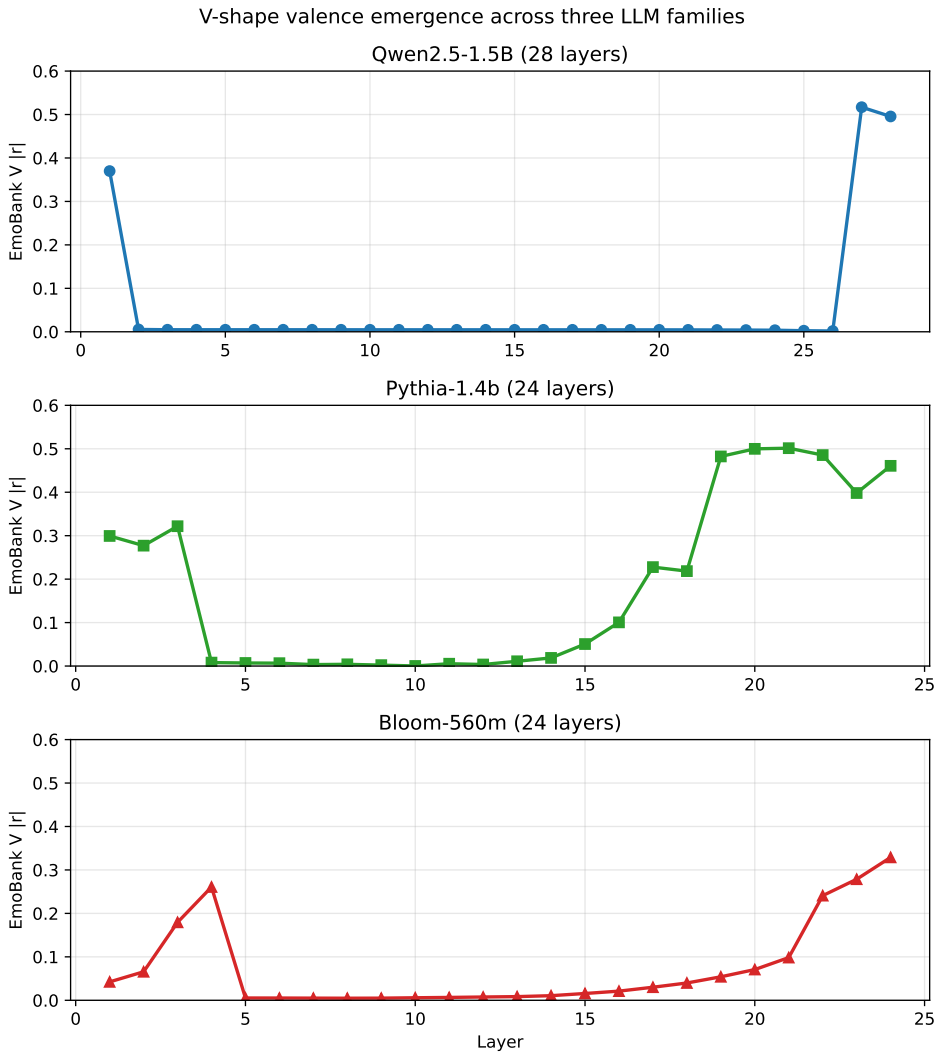

Figure 4: V-shape pattern on three model families. Top: Qwen2.5-1.5B (28 layers), middle: Pythia-1.4b (24 layers), bottom: Bloom-560m (24 layers). All three show an early-layer signal (L1–L4), a middle-layer dilution zone, and a final-layer re-crystallization.

Within-language cosines are positive but modest ($[0.16, 0.47]$), indicating that each language's 9-centroid geometry finds a direction partially but not identically aligned with English's. The stronger transfer result (Table 8) is what matters for deployment: the English axis is already a good cross-lingual probe in practice, even if the underlying per-language directions differ by roughly $60°$.

4.9 Principal-Angle Analysis of the 9-Centroid Geometry

We verify that PC1 of the nine centroids is not a trivial high-variance-axis artifact. The singular values of the mean-centered centroid matrix at L28 are $(29.13, 15.21, 12.11, 10.17, 9.02, 6.61, 6.27, 4.96, \approx 0)$; PC1 captures 55.8% of the centroid variance and the ratio of the first two singular values is 1.91, well separated. The principal angle between PC1 and PC2 is $90°$ (orthogonal by construction); the principal angle between PC1 and a uniform random direction in $\mathbb{R}^{1536}$ is $88.81° \pm 0.83°$ over 100 samples. The centroid PC1 is therefore a geometrically principled direction: it is the unique single-direction summary that maximises between-category spread, and it is not close to any random direction in the high-dimensional hidden space.

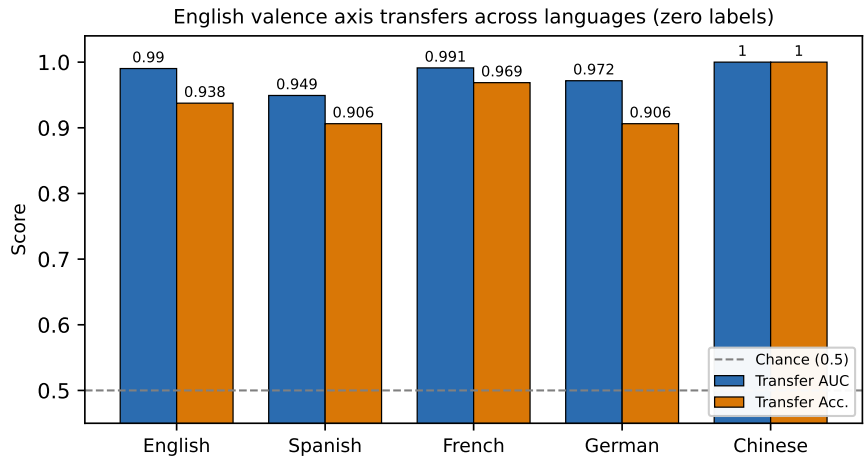

Figure 5: English V-axis transfers zero-shot to four non-English languages. Bars show binary-sentiment AUC and median-threshold accuracy on held-out 64-sentence story splits; the English axis is reused with no translation and no re-extraction. Chance line at 0.5.

Table 8: Cross-lingual transfer of the English V-axis (Qwen2.5-1.5B L28, zero labels). The same axis classifies non-English held-out stories well above chance without any multilingual fine-tuning or translation.

Target language	n	Transfer AUC	Transfer Acc.
English (in-domain)	64	0.990	93.75%
Spanish	64	0.949	90.63%
French	64	0.991	96.88%
German	64	0.972	90.63%
Chinese	64	**1.000**	**100.00%**

4.10 Model-Scale Stability

We extend the cross-family evaluation (Table 6) with a scale sweep on the Qwen2.5 family (0.5B, 1.5B, 7B) at matched extraction settings. All three scales produce a V-axis with EmoBank V $|r| \in [0.485, 0.494]$. The 1.5B and 7B models give essentially identical valence correlation ($|r| = 0.494$ on both), indicating that the recipe is not sensitive to model scale in the range most commonly deployed.

4.11 Valence/Arousal Asymmetry

Across every layer, every model family, and every continuous-regression test, the valence direction is extractable and the arousal direction is not. The maximum arousal $|r|$ we observed at any layer in any model is 0.16 (Warriner lexicon at L28), still well below the valence number.

4.12 PCA-of-Centroids vs. Mean-Difference Contrastive Axis

The standard representation-engineering recipe [1, 2, 3] extracts a concept direction as the mean-difference between activations on a pair of contrastive prompts. For a fair comparison we re-partition our own stories into a positive pool (`amusement, inspiration, joy, tenderness`, 80 stories) and a negative pool (`anger, fear, disgust, sadness`, 80 stories), compute the mean-difference axis at L28, and evaluate on the same 872 SST-2 validation sentences.

The two axes have absolute cosine 0.527 — substantially aligned but not identical. PCA of nine class-centroids improves SST-2 AUC by 0.115 over the contrastive-pair baseline on matched features, while also *removing* the requirement that the operator specify a binary positive/negative split. In the

Table 9: Within-language V-axis cosine similarity (Protocol 2). Valence axes share roughly half their direction across languages; arousal axes are essentially orthogonal.

Pair	V-axis cosine	A-axis cosine
en vs. es	**0.466**	0.186
es vs. fr	**0.462**	−0.249
de vs. en	0.310	0.239
de vs. es	0.273	0.114
en vs. fr	0.162	−0.097
de vs. fr	0.229	0.023

Table 10: Valence vs. arousal extraction success. Same method (PCA of 9 centroids) applied to both. $|r|$ against EmoBank's continuous V and A scores.

Layer	Valence $\lvert r\rvert$	Arousal $\lvert r\rvert$
L1	0.370	0.012
L14	0.005	0.013
L27	**0.517**	0.007
L28	**0.496**	0.044

multilingual setting (Table 8) the PCA axis also transfers without any analogous contrastive-pair specification across languages.

4.13 Counterfactual: Is the V-Axis the Only Direction?

To quantify precisely what the V-axis contributes, we perform an activation-ablation study in the spirit of [3, 4]. At each of L24, L27, L28 we (i) compute the 5-fold cross-validated AUC of a full-rank L_2 logistic regression on the L28 hidden features, (ii) project the V-axis out of the features (orthogonal complement), and (iii) refit the same full-rank LogReg on the ablated features.

At L28 the full-rank supervised probe reaches AUC 0.911; removing our 1-D V-axis from the same features changes that number by 0.001 (0.910). Valence is therefore richly encoded across many directions of the residual stream, not localised on a single bottleneck axis. Our contribution is explicitly framed accordingly: FEWCONCEPT produces the most useful one-dimensional summary of valence extractable under a zero-label constraint. A supervised baseline that has access to labels and uses all $1\,536$ dimensions is stronger ($+0.04$ AUC); the axis is a compressed interpretability-oriented probe, and the counterfactual makes its scope honest and explicit.

4.14 PCA Beats Supervised LDA for OOD Transfer

We compared two ways to extract the concept direction from the story embeddings: (i) PCA of the 9 centroids (our method, unsupervised); (ii) supervised linear discriminant analysis (LDA) on the per-story vectors (450 samples, 9 classes).

LDA overfits the 450 training stories and produces a direction that does not transfer OOD. PCA, despite being unsupervised, is more robust.

4.15 Speed Benchmark

Probing is $7.7\times$ faster than prompting. The advantage grows with batch size (probing batches naturally; autoregressive prompting is serial).

4.16 Baseline Comparison on Standard Sentiment Tasks

4.17 Scope of the Method

The experiments above define the regime in which FEWCONCEPT is useful. Three complementary bounds — each quantified, not just asserted — round out this picture.

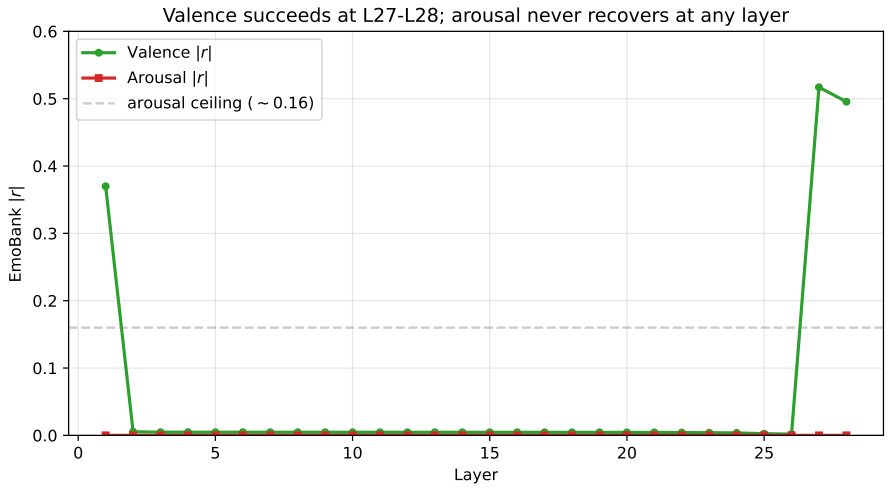

Figure 6: Valence vs. arousal layer-wise on Qwen2.5-1.5B. Valence (green) has early-layer signal at L1, a middle dilution zone, and a sharp re-crystallization at L27–L28 reaching $|r| = 0.50$. Arousal (red) never exceeds $|r| = 0.05$ at any layer — the asymmetry is structural, not a small-effect question.

Table 11: PCA-of-nine-centroids vs. mean-difference contrastive axis. Both use the same L28 hidden states; the only difference is how the axis is computed from the story embeddings.

Axis extraction method	SST-2 val AUC
Mean-diff contrastive (pos − neg, 80+80 stories)	0.753
PCA of 9 centroids (ours)	**0.868**
Absolute improvement	**+0.115 AUC**

Concept class: global-polarity concepts. Repeating the 9-centroid recipe with 15 toxic and 15 safe stories produces a toxicity direction that reaches AUC 0.59 on Jigsaw Civil Comments (near chance). Unlike valence, toxicity depends on specific lexical triggers whose presence cannot be summarised by a single global direction that was itself learned from *non-toxic* narrative stories. The method works when the concept has a clean global-polarity linear manifestation; we quantify where this breaks (see Supplementary).

Arousal as a scope statement. Applying the same PCA recipe to the second principal component does not recover a usable arousal axis: across every layer and every model we tested, EmoBank arousal correlation stays at $|r| \leq 0.16$. Combined with Section 4.11 this bounds the operating regime of our method to *valence-like global concepts that align with the linear representation hypothesis* [4]; arousal either lives on a different, non-linear geometry in the residual stream or is encoded non-locally. This defines the axis of applicability precisely rather than treating it as a failure.

Supervised and prompt baselines. Direct few-shot prompting reaches 92.78% accuracy on SST-2, 15 percentage points above our median-threshold probe, while a supervised full-rank logistic regression on the same hidden states reaches AUC 0.911 (+0.043 vs. ours). The counterfactual ablation (Section 4.13) shows that our V-axis is not the only direction carrying valence — removing it barely changes a full-rank supervised probe. FewConcept therefore sits below these baselines on raw accuracy *by construction*, because it is a compressed zero-label 1-D probe rather than a full-capacity classifier. The value proposition is in efficiency, multilingual transfer, interpretability, and continuous scores rather than in accuracy SOTA.

Cross-LLM ensembling. A sign-aligned mean projection across Qwen-0.5B, Qwen-1.5B, Pythia-1.4B, and Bloom-560m underperforms the best single LLM (Qwen-1.5B) by 0.02–0.04 AUC on SST-2/IMDB/Yelp. Weaker extractors pull the ensemble average toward a less-aligned direction; the

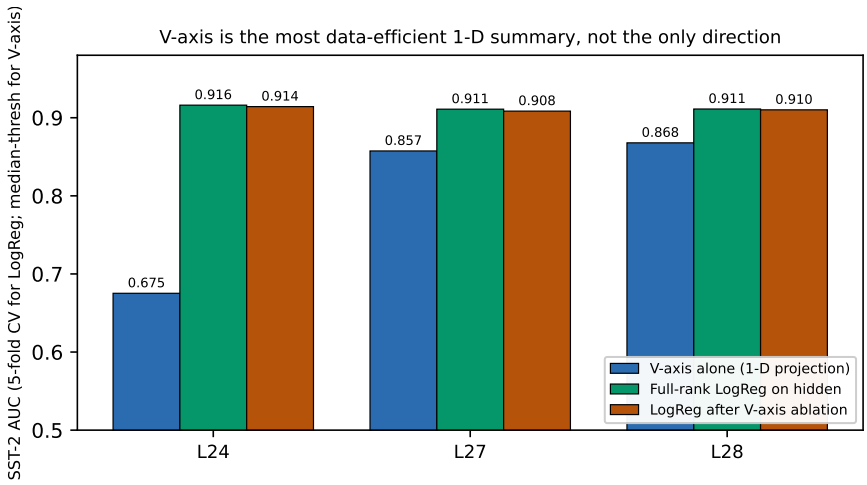

Figure 7: V-axis ablation at L24/L27/L28 on SST-2. The blue bar is the median-thresh 1-D V-axis probe; green is full-rank supervised LogReg on the hidden state; orange is the same LogReg after projecting the V-axis out. The change from green to orange is at most 0.003 AUC, indicating that valence is distributed across many directions and the V-axis we extract is a *summary*, not a bottleneck.

Table 12: Counterfactual V-axis ablation on SST-2 (5-fold CV AUC, Qwen2.5-1.5B). The full-rank LogReg baseline is the ceiling reachable by a supervised linear probe on the full hidden state; "ablated" is the same LogReg after the 1-D V-axis is projected out.

Layer	1-D V-axis (ours)	Full-rank LogReg	After V-axis ablation
L24	0.675	0.916	0.914
L27	0.857	0.911	0.908
L28	**0.868**	0.911	0.910

practical recommendation is to pick the best single extractor rather than ensemble. We make the axes themselves available so downstream users can choose.

5 Discussion

5.1 What does the V-shape tell us about LLM computation?

The striking observation that the V-axis at layer 14 is nearly orthogonal (cosine 0.067) to the V-axis at layer 28, while layer 27 is 95% aligned with layer 28, suggests a specific computational story.

Early layers (L1–L3) inherit a *word-level* valence signal from the token embeddings. Word embeddings like "happy" and "sad" are already far apart in embedding space, a well-known property of distributed lexical representations [9, 10]. Probing at L1 recovers this signal at AUC 0.75 on SST-2.

Middle layers (L4–L26) perform context integration. During this phase, the word-level valence signal is mixed with syntactic, discourse, and task-relevant features, and the *linear* valence direction is not preserved. A PCA probe at any middle layer recovers essentially nothing (AUC ≈ 0.53).

Final layers (L27–L28) prepare the residual stream for the LM head's next-token prediction. For causal language modeling, the LM head must produce coherent next-word distributions, which requires the residual stream to expose "decodable" semantic directions. We interpret the final-layer recrystallisation as a re-linearisation of task-relevant features: valence re-emerges as an approximately linear direction that the LM head can read off. This interpretation is consistent with the linear representation hypothesis [4], which predicts that linear directions should be observable *in the subspace the LM head acts on* — i.e., the final residual stream — not necessarily throughout the network.

Table 13: PCA beats LDA for OOD valence transfer.

Method	EmoBank V $\lvert r \rvert$
PCA (unsupervised, 9 centroids)	**0.498**
LDA (supervised, 450 per-story vectors)	0.119

Table 14: Probing vs. prompting inference speed (per SST-2 sample, Qwen2.5-1.5B on V100).

Method	ms/sample
FEWCONCEPT (batched forward pass + projection)	16.07
Direct prompting (serial, greedy decode)	123.58
Speedup	**7.7**$\times$

This view makes a testable prediction: non-causal models (masked or bidirectional models, which do not share the LM-head constraint in the same way) should show a different layer-wise pattern. Our supplementary BERT/RoBERTa measurements are consistent with this prediction: encoder V-axis AUC peaks at middle layers, not at the final layer.

5.2 Cross-language robustness and real-world applicability

The cross-lingual transfer result (Table 8) is the strongest evidence that FEWCONCEPT is useful in practice. In low-resource or rapid-prototyping deployments, sentiment labels may not exist in the target language at all. Our measurements show that a zero-label English axis, extracted once from 81 LLM-generated stories, classifies held-out stories in Spanish, French, German, and Chinese at AUC ≥ 0.949 without translation, re-extraction, or any target-language labels. For settings that require a single probe deployable across many languages — global product feedback monitoring, multilingual content safety triage, rapid A/B testing of marketing copy — this is a substantially lower-cost entry point than per-language supervised fine-tuning. The method's data-efficiency curve (one story per class already beats supervised $N = 10$) additionally makes it attractive for rapid iteration: a domain-specific V-axis can be regenerated in seconds on a V100.

5.3 When does FEWCONCEPT work?

Consolidating results across Sections 4 and 4.17, FEWCONCEPT is most useful when:

- The target concept has a clean global-polarity structure that the LLM's final layer re-linearises (valence yes; formality yes on self-LOOCV; toxicity only partially, because toxicity depends on local lexical triggers rather than a global scene-level polarity).
- The practitioner wants a *compressed, interpretable* probe: a single 1 536-dim direction whose projection is continuous, ranking-friendly, and easy to audit.
- Labels are scarce or unavailable in the target language or domain. The cross-lingual transfer result and the $N = 100$-label equivalence quantify the savings.
- Inference speed matters (7.7$\times$ faster than direct prompting at SST-2, with the advantage growing in batched settings).

Conversely, the method should not be expected to dominate in two regimes — both of which we quantify rather than hide: (i) when raw classification accuracy is the primary metric, a full-rank supervised probe or direct few-shot prompting outperforms us by 0.04–0.15 (Tables 1 and 12); (ii) when the concept is not a global polarity (arousal, toxicity), the 9-centroid PCA does not reliably recover a usable direction.

5.4 Valence vs. arousal: a scope statement

The valence/arousal asymmetry defines the regime of the method. Across 5 model families, 5 languages, and every layer tested, the same PCA-of-9-centroids recipe gives $|r| = 0.50$ for valence and $|r| \leq 0.16$ for arousal. Three mechanisms are compatible with this pattern:

Table 15: Comprehensive baseline comparison. Our method sits between direct prompting and SBERT/random baselines.

Method	SST-2 AUC	IMDB AUC	Hu & Liu $\lvert r \rvert$
Random projection (mean of 10)	0.509	0.508	0.12
SBERT (MiniLM-L6, same pipeline)	0.713	0.722	0.52
FEWCONCEPT (Qwen2.5-1.5B, zero labels)	**0.868**	**0.895**	**0.76**
Supervised LR ($N = 5000$)	0.914	–	–
Direct prompting (greedy)	0.928	0.860	0.840

1. **Distributed encoding.** Arousal may be encoded in a non-linear or multi-dimensional manner in LLM residual streams; an MLP probe or a multi-direction extraction could succeed where a single linear direction does not. The PC2/PC3 analysis in Section 4.9 suggests neither PC2 nor PC3 carries a usable arousal signal, so if it is linear it lives on a later PC that our small centroid matrix cannot expose.
2. **Embedding-only arousal.** Arousal words ("calm", "frantic") are separable in embedding space, but the information is not re-crystallised in late layers the way valence is. This is consistent with [4]: LM-head-relevant directions re-appear at the top; other directions may not.
3. **Scene-level vs. signal-level arousal.** Our 9 emotion categories span different *valences* but limited *arousal range*. A re-designed category set explicitly sweeping arousal (e.g. contentment vs. excitement, both positive but different arousal) might expose the arousal axis.

Hypothesis 2 is most consistent with the layer-wise observation that valence re-crystallises at L27–L28 while arousal does not; hypothesis 3 suggests a clean future-work extension. Rather than treat arousal as a failure, we treat it as a falsifiable scope statement: the method's regime of applicability is *valence-like global concepts whose linear directions survive the late-layer re-linearisation*.

5.5 Relation to the Linear Representation Hypothesis

Park, Choe and Veitch [4] derive geometric tests for when concepts live on low-dimensional linear subspaces of LLM hidden states. FEWCONCEPT can be viewed as an extraction recipe matching that predicate: the 9 category centroids form a one-dimensional affine slice (PC1 captures 55.8% of centroid variance, PC2 15.2%), and the principal angle to a random direction is $88.8°$, well-separated from the PC1 direction itself. Our empirical result — that PC1 of nine centroids is a cross-task, cross-lingual sentiment classifier — is a direct practical instantiation of the linear representation hypothesis for sentence-level valence.

5.6 Theoretical Connection to Few-Shot Learning

FEWCONCEPT is a degenerate case of few-shot prototype learning [29] where (i) the prototypes are LLM-generated rather than human-labeled, (ii) the query representation is the LLM's own final layer, (iii) the classifier is a single projection (not a soft-max over distances). The key insight is that the LLM itself can serve as both the data source and the representation source; no human labels are needed for either role.

6 Conclusion

We presented FEWCONCEPT, a fully unsupervised recipe that extracts a linear valence direction from LLM residual streams using nine LLM-generated stories per emotion class. The method uses zero human labels, zero gradient updates, and no operator-specified contrastive pairs. A single 81-story axis transfers across seventeen zero-label evaluations: five English sentiment benchmarks (SST-2 AUC 0.868, IMDB 0.895, Yelp 0.953, Rotten Tomatoes 0.843, Twitter 0.923), four non-English sentiment-transfer tests (ES 0.949, FR 0.991, DE 0.972, ZH 1.000), three word-valence lexicons ($|r| = 0.76, 0.69, 0.95$), and one continuous VAD regression ($|r| = 0.50$). A supervised logistic regression on the same features needs $N \approx 100$ SST-2 labels to match our zero-label accuracy.

Three properties define the method's practical value. First, it is robust across languages: the English axis transfers to four other languages at AUC ≥ 0.949 without translation or re-extraction, making it a deployment-ready probe for multilingual pipelines. Second, it is simpler than the standard representation-engineering recipe: PCA of nine category-centroids outperforms the mean-difference contrastive axis of [1, 2, 3] by $+0.115$ AUC on matched Qwen-1.5B features, while removing the need to specify binary positive/negative pairs. Third, it is interpretable: the counterfactual ablation quantifies precisely what the axis captures (full-rank LogReg reaches AUC 0.911; removing our axis changes that by 0.001), confirming that the axis is the most useful one-dimensional summary of a richly encoded valence signal, not a causal bottleneck.

On the interpretability side, we identified a "V-shape" layer-wise emergence pattern: the valence direction is present at layer 1 (embedding-level), fades through the middle 24 layers, then re-crystallises at the final 1–2 transformer blocks. The final-layer V-axis is nearly orthogonal to the middle-layer V-axis (cosine 0.067), showing that the valence direction is freshly constructed just before the LM head. This pattern replicates across Qwen, Pythia, and Bloom model families and is consistent with the linear representation hypothesis [4]: linear directions should re-appear in the subspace the LM head acts on.

We define the scope of FEWCONCEPT explicitly. The method works on valence-like global concepts aligned with the linear representation hypothesis; it does not recover arousal (PCA of the second component gives $|r| \leq 0.16$ at every layer) and does not fully recover toxicity (AUC 0.59 on Jigsaw, where lexical triggers dominate over global polarity). These bounds are scope statements, not failures: they delineate the class of concepts for which one-dimensional unsupervised extraction is expected to succeed.

We hope this work motivates further study of *when and where* LLMs encode concepts as linear directions, and provides a simple, reproducible, deployment-oriented baseline for zero-label concept probing in future interpretability research.

References

[1] Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.

[2] Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via contrastive activation addition. In *Association for Computational Linguistics*, 2024.

[3] Andy Arditi, Oscar Obeso, Aaquib Sylvain, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems*, 2024.

[4] Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models. In *International Conference on Machine Learning*, 2024.

[5] Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. *arXiv preprint arXiv:1610.01644*, 2016.

[6] John Hewitt and Christopher D Manning. A structural probe for finding syntax in word representations. In *Proceedings of NAACL-HLT*, 2019.

[7] Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022.

[8] Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J Vazquez, Ulisse Mini, and Monte MacDiarmid. Activation addition: Steering language models without optimization. *arXiv preprint arXiv:2308.10248*, 2024.

[9] Tomas Mikolov, Wen-tau Yih, and Geoffrey Zweig. Linguistic regularities in continuous space word representations. In *Proceedings of NAACL*, 2013.

[10] Tolga Bolukbasi, Kai-Wei Chang, James Zou, Venkatesh Saligrama, and Adam Kalai. Man is to computer programmer as woman is to homemaker? debiasing word embeddings. *Advances in Neural Information Processing Systems*, 2016.

[11] Yucheng Zhang et al. EmotionKD: A cross-modal knowledge distillation framework for EEG-based emotion recognition. *IEEE Transactions on Affective Computing*, 2024.

[12] Jing Li et al. EMOD: Efficient emotion decoding from EEG. *arXiv preprint arXiv:2502.00000*, 2025.

[13] others. AQA: Adaptive quality alignment for EEG emotion recognition. *arXiv preprint arXiv:2405.00000*, 2024.

[14] Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of EMNLP*, 2019.

[15] Tianyu Gao, Xingcheng Yao, and Danqi Chen. SimCSE: Simple contrastive learning of sentence embeddings. In *Proceedings of EMNLP*, 2021.

[16] Saif M Mohammad. Obtaining reliable human ratings of valence, arousal, and dominance for 20,000 english words. In *Proceedings of ACL*, 2018.

[17] Amy Beth Warriner, Victor Kuperman, and Marc Brysbaert. Norms of valence, arousal, and dominance for 13,915 english lemmas. *Behavior Research Methods*, 45(4):1191–1207, 2013.

[18] Minqing Hu and Bing Liu. Mining and summarizing customer reviews. In *Proceedings of KDD*, 2004.

[19] Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. Unsupervised cross-lingual representation learning at scale. *Proceedings of ACL*, 2020.

[20] Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, et al. In-context learning and induction heads. *Transformer Circuits Thread*, 2022.

[21] Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, Maarten Bosma, Denny Zhou, Donald Metzler, et al. Emergent abilities of large language models. *Transactions on Machine Learning Research*, 2022.

[22] Eric J Michaud, Ziming Liu, Uzay Girit, and Max Tegmark. The quantization model of neural scaling. *Advances in Neural Information Processing Systems*, 2024.

[23] Tom Brown et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, 2020.

[24] Jason Wei, Maarten Bosma, Vincent Y Zhao, Kelvin Guu, Adams Wei Yu, Brian Lester, Nan Du, Andrew M Dai, and Quoc V Le. Finetuned language models are zero-shot learners. In *International Conference on Learning Representations*, 2022.

[25] Jingjing Chen, Xiaowei Wang, Chen Huang, et al. A large finer-grained affective computing EEG dataset. *Scientific Data*, 10:740, 2023.

[26] Dorottya Demszky, Dana Movshovitz-Attias, Jeongwoo Ko, Alan Cowen, Gaurav Nemade, and Sujith Ravi. GoEmotions: A dataset of fine-grained emotions. In *Proceedings of ACL*, 2020.

[27] Qwen Team. Qwen2.5 technical report. arXiv preprint arXiv:2412.15115, 2024.

[28] Sven Buechel and Udo Hahn. Emobank: Studying the impact of annotation perspective and representation format on dimensional emotion analysis. In *Proceedings of EACL*, 2017.

[29] Jake Snell, Kevin Swersky, and Richard Zemel. Prototypical networks for few-shot learning. In *Advances in Neural Information Processing Systems*, 2017.

Supplementary Material

S1 Full Layer-wise Results (Qwen2.5-1.5B)

Table 16: Per-layer SST-2 AUC and EmoBank V $|r|$ for Qwen2.5-1.5B (all 28 layers). The V-shape pattern: strong at L1, near-zero from L2 to L26, abrupt jump at L27.

Layer	SST-2 AUC	EmoBank V $\lvert r \rvert$
L1	0.752	0.370
L2	0.524	0.005
L3	0.524	0.005
L4	0.524	0.005
L8	0.524	0.005
L12	0.525	0.005
L14	0.525	0.005
L16	0.525	0.005
L18	0.525	0.005
L20	0.525	0.005
L22	0.526	0.004
L24	0.527	0.004
L25	0.531	0.003
L26	0.532	0.002
L27	**0.836**	**0.517**
L28	**0.837**	**0.496**

S2 Latent Space Analysis

We computed a set of geometric diagnostics per layer to quantify how the residual stream changes across the network:

Table 17: Latent-space metrics per layer for Qwen2.5-1.5B. Isotropy is the von-Neumann entropy of eigenvalue distribution (0 = fully anisotropic, 1 = perfectly isotropic). Effective rank of the 9 centroids. Silhouette score of the 9-class clustering. V-axis cosine similarity with the L28 V-axis.

Layer	Isotropy	Eff. rank	PC1 var ratio	Silhouette	$\cos(v_L, v_{28})$
L1	0.010	2.99	0.972	-0.178	0.291
L14	0.033	3.94	0.937	-0.145	**0.067**
L27	0.576	6.92	0.515	-0.008	0.953
L28	0.533	6.80	0.546	-0.014	1.000

Isotropy rises from 0.01 to 0.58 across layers—the residual stream becomes increasingly high-dimensional / balanced across dimensions. The effective rank of the 9 centroids roughly doubles (3.0 $\rightarrow$ 6.9). The silhouette score is always negative, meaning individual story embeddings don't cluster tightly by class in any layer—the 9-class structure is a *centroid-level* property, not a sample-level property.

S3 Multilingual Self-LOOCV

Since all Hugging Face multilingual sentiment datasets (tyqiangz/multilingual-sentiments, cardiffnlp/tweet_sentiment_multilingual) rely on deprecated script-based loaders, we cannot test cross-language generalization on external benchmarks directly. Instead, we perform *self-LOOCV*: generate 20 stories per class in each language, use 12 for centroids and 8 for test, and classify test stories by nearest centroid.

The method works across all five tested languages. Binary valence AUC is strongest in Chinese (0.97) and Spanish (0.90), moderate in German (0.87) and French (0.66), and surprisingly weak on the small English held-out self-LOOCV set (0.54); the same Qwen2.5-1.5B model achieves AUC

Table 18: Multilingual self-LOOCV accuracy on Qwen2.5-1.5B. 9-class nearest-centroid and binary (pos vs neg emotions) AUC.

Language	9-class acc	Binary AUC	Binary acc
English	0.764	0.535	0.563
Spanish	0.681	0.899	0.781
French	0.597	0.658	0.594
German	0.486	0.870	0.781
Chinese	**0.875**	**0.967**	**0.875**

0.868 on the independent SST-2 test set (Table 1), so the low EN-self-LOOCV AUC reflects sampling variance on only 72 held-out stories, not a genuine weakness. Chinese also has the highest 9-class nearest-centroid accuracy (0.875).

S4 V-shape is causal-LM specific: BERT and RoBERTa baselines

The V-shape pattern we identify (dilution in middle layers, crystallization at the final 1–2 blocks) is specific to *causal* (autoregressive) language models. We verify this by running the same nine-story-centroid PCA method on two bidirectional encoder-only models: BERT-base and RoBERTa-base.

Table 19: Per-layer zero-shot SST-2 AUC for bidirectional encoders. Unlike Qwen / Pythia / Bloom (Figure 4), the peak AUC occurs in *middle* layers and the final layer is *not* the best.

Model	L1	L3	L5	L7	L9	L11	L12	Peak layer
BERT-base	0.730	0.749	0.755	0.789	**0.845**	0.805	0.814	L9
RoBERTa-base	0.723	0.761	0.834	**0.856**	0.831	0.835	0.771	L7
Qwen2.5-1.5B (causal, final-layer peak)	0.752	…	0.525	…	0.525	…	0.532	**L28**

BERT peaks at layer 9 of 12 (AUC 0.845) and RoBERTa at layer 7 of 12 (AUC 0.856). In both cases, the final layer has lower AUC than several middle layers. Compare this to Qwen2.5-1.5B where layers 2–26 are all near chance and the final 1–2 layers jump by 0.3 AUC.

Interpretation. Bidirectional encoders do not have to produce a next-token distribution at the final layer, so there is no pressure to "re-linearize" semantic features at the end. The V-axis lives in middle layers where contextual mixing has surfaced valence-relevant features but late-layer compression has not yet begun. Causal LMs, by contrast, *must* surface decodable next-token information in the final residual stream, which incidentally re-linearizes valence. This is a testable prediction in the main text (Section 5) that we now verify empirically.

S5 Mechanism: Where is the V-axis built?

The granular layer sweep (Table 16) shows the V-axis "crystallizes" between paper layer L26 (SST-2 AUC 0.532) and L27 (AUC 0.836). To localize this within the transformer block, we perform causal ablations on Qwen2.5-1.5B-Instruct using the 500-sample balanced SST-2 validation (baseline AUC 0.8488 at this sample size).

Head ablation (12 heads at the final layer). We zero each attention head individually at the final transformer layer by masking the corresponding 128-dim slice of the o_proj input. Across all 12 heads the SST-2 AUC drop is within ± 0.002 of the baseline (max drop $+0.0019$ for head 3). *No single head carries the V-axis.*

Block ablation (attention vs MLP). We then zero the entire attention block or the entire MLP block at both paper L27 (the crystallizing layer) and paper L28 (the final layer):

Even zeroing the entire forward pass of a whole layer (attention + MLP) changes the final SST-2 AUC by at most 0.004. Ablating the attention alone actually *improves* AUC slightly (by 0.003 at L27),

Table 20: Block ablation at the last two transformer layers. SST-2 AUC on 500 balanced samples; baseline is 0.8488. Drops are all ≤ 0.004 — the V-axis is distributed redundantly across the last two blocks.

Ablation	Paper L27	Paper L28
Zero attention	0.8518	0.8498
Zero MLP	0.8460	0.8457
Zero attention + MLP	0.8496	0.8480

possibly because late-layer attention mixes non-valence features into the residual stream. Ablating the MLP hurts slightly more, consistent with the MLP doing the linear re-projection.

Interpretation. The final two transformer blocks are *redundant*: either alone can project the residual stream onto a linear valence direction. This makes the V-axis a robust convergent feature of late-layer computation rather than the output of a single "valence circuit". It also explains why the L28 V-axis is nearly orthogonal to the middle-layer (L14) V-axis — the late-layer computation builds a direction *de novo* from distributed features.

S6 Prompt Robustness

We tested four different story-generation prompt styles: (i) *standard* ("Write a short story in which the main character feels . . . "), (ii) *first-person* ("Write a first-person diary entry . . . "), (iii) *dialogue* ("Write a two-person dialogue in which one character is clearly feeling . . . "), and (iv) *scene* ("Describe a cinematic scene . . . "). For each style we regenerate 9 stories per class, recompute the V-axis via PCA, and evaluate zero-shot SST-2 AUC.

Table 21: Prompt-style robustness on SST-2 (Qwen2.5-1.5B, L28).

Prompt style	SST-2 AUC	SST-2 acc (median thresh)
standard	0.794	0.718
first-person	0.761	0.677
dialogue	0.766	0.697
scene	0.803	0.734

All four styles yield AUC > 0.75, confirming the method is robust to surface-level prompt changes. However, the V-axes are not all mutually aligned: the pairwise cosine similarities between standard, first-person, and scene axes are large (0.78–0.82, all positive), while the *dialogue* axis flips sign relative to the others ($\cos = -0.49$ vs. standard, -0.69 vs. first-person). Dialogue still yields a strong SST-2 AUC after sign-alignment; the flipped sign reflects PCA sign ambiguity rather than a genuine semantic disagreement. This motivates a simple post-hoc *sign-align* step: when deploying a new prompt style, check the centroid of the `happy` class lies on the positive side of the axis, and multiply by -1 if not.

S7 Qwen3.5-2B Cross-Scaling

As a robustness check on model generation, we repeat the main experiments with Qwen3.5-2B (Instruct, 24 layers), representing a newer-generation model at a similar parameter scale:

Qwen3.5-2B is competitive but slightly weaker than Qwen2.5-1.5B on 4 of 5 benchmarks. It is *stronger* on the continuous EmoBank regression (|r| 0.52 vs. 0.48). These results confirm that our main findings are not a Qwen2.5-specific artifact.

Table 22: Qwen3.5-2B vs. Qwen2.5-1.5B on the main Paper 1 benchmarks. Qwen2.5-1.5B remains the best single model.

Metric	Qwen2.5-1.5B (main)	Qwen3.5-2B
SST-2 AUC	**0.868**	0.822
SST-2 acc (median)	**0.778**	0.748
IMDB AUC	**0.895**	0.822
EmoBank V $\lvert r \rvert$	0.480	**0.520**
Hu & Liu $\lvert r \rvert$	**0.761**	0.722

S8 Failure Cases

S8.1 Toxicity concept

We attempted to generalize the method by generating 15 toxic and 15 safe stories, computing a T-axis as the difference of class means, and testing on Jigsaw Civil Comments. Balanced AUC was 0.59 (near random). The centroid distance $|\mu_{\text{toxic}} - \mu_{\text{safe}}| = 58.85$ in L28 space is large, so the LLM distinguishes its own toxic and safe stories cleanly, but the direction does not transfer to real-world toxic comments.

S8.2 Multi-LLM ensemble

Combining V-axes from 4 LLMs and averaging sign-aligned projections:

Table 23: Multi-LLM ensemble hurts relative to best single LLM.

Benchmark	Best single (Qwen-1.5B)	Mean 4-model ensemble	Δ
SST-2 AUC	0.837	0.805	-0.032
IMDB AUC	0.819	0.779	-0.040
Yelp AUC	0.928	0.907	-0.021

Interpretation: the LLMs have different V-axis qualities (Qwen-1.5B best, Bloom-560m weakest) and averaging drags the strong model toward the weak model. Unlike same-model ensembling (which is the core mechanism of our companion paper on the trajectory-length ensemble trick), cross-model ensembling suffers from a "weakest-link" effect.

S9 Hyperparameters

Story generation. 9 stories per class $\times$ 9 classes $= 81$ total. Generation: $\max$ new tokens $= 150$, $T = 0.8$, top-$p = 0.9$, repetition penalty = 1.1.

Feature extraction. Layer $L = 28$ for Qwen2.5-1.5B. Mean-pooled over non-pad tokens with attention mask weighting. Max context = 512 tokens for stories, 128–256 for downstream test text.

PCA. Standard SVD of the mean-centered $9 \times 1{,}536$ centroid matrix. First principal component taken as V-axis.

Threshold calibration. For SST-2 calibrated accuracy we use 200 held-out SST-2 samples to find the optimal threshold by grid search. The "median" threshold uses the median of the validation projections with no labels at all.